

Wind Driven Ocean Gyres

Sverdrup and Munk Solutions

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Introduction

Ocean circulation gyres are part of the atmospheric circulation.

- 1 Heating air at the equator → Hadley cells.
- 2 Coriolis effect deflects the air movement → Trade winds and westerlies
- 3 Enforce wind stress on the ocean surface → 90° Ekman transport.
- 4 Since f increases with latitude, net southward Ekman transport in the North Atlantic.
- 5 Western boundary current needed to close the circulation.



Figure: Wikimedia Commons, North Atlantic location map, accessed 2026

Introduction, Sverdrup

Introduction

Theory

Sverdrup

Munk

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Result

The Sverdrup
solution

The Munk
solution

Discussion

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References

Harald Ulrik Sverdrup (1888-1957) formulated the net transport in the ocean interior as:

$$\boxed{-\beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau}} \quad (1)$$

which gives the Sverdrup solution:

$$\begin{aligned} \int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x} dx &= \int_x^{x_{\text{East}}} vH dx = \cancel{\Psi(x_{\text{East}}, y)} - \Psi(x, y) \\ \Rightarrow \Psi(x, y) &= -\frac{1}{\rho\beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau}) \end{aligned}$$

Introduction, Munk

Walter Heinrich Munk (1917-2019) extended the Sverdrup model by including meridional harmonic Newtonian viscosity (diffusion of momentum):

$$A \nabla^4 \Psi - \beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau} \quad (2)$$

which gives the Munk solution:

$$\Psi(x, y) = \Psi_{\text{Sverdrup}}(x, y) \left(1 - e^{-\frac{x - x_{\text{West}}}{2\delta_m}} \left[\cos\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) + \frac{1}{\sqrt{3}} \sin\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) \right] \right)$$

$$\text{where } \delta_M \equiv \left(\frac{A}{\beta} \right)^{\frac{1}{3}}$$

Stress set as:

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin \left(\frac{\pi y}{L_y} \right). \quad (3)$$

<i>Variable</i>	<i>Sverdrup model</i>	<i>Munk model</i>
Atlantic, L_x, L_y	7000 km, 7000 km	7000 km, 7000 km
Pacific, L_x, L_y	–	14000 km, 7000 km
Grid, $\Delta x, \Delta y$	35 km	35 km
Wind forcing, U_{10}	[4, 5, 6] m/s	5 m/s
Viscosity, A	0	$[3 \cdot 10^4, 10^5, 5 \cdot 10^5] \text{ m}^2/\text{s}$
$\delta_M / \Delta x$	–	[3.3, 4.9, 8.4]

Table: Comparison of parameters used in the Sverdrup and Munk models.

Result, The Sverdrup solution

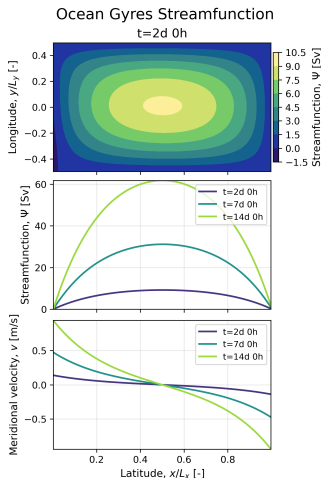


Figure: Streamfunction for the no-drag f -plane configuration. The circulation remains symmetric with no β -effect and without western intensification.

Result, The Sverdrup solution

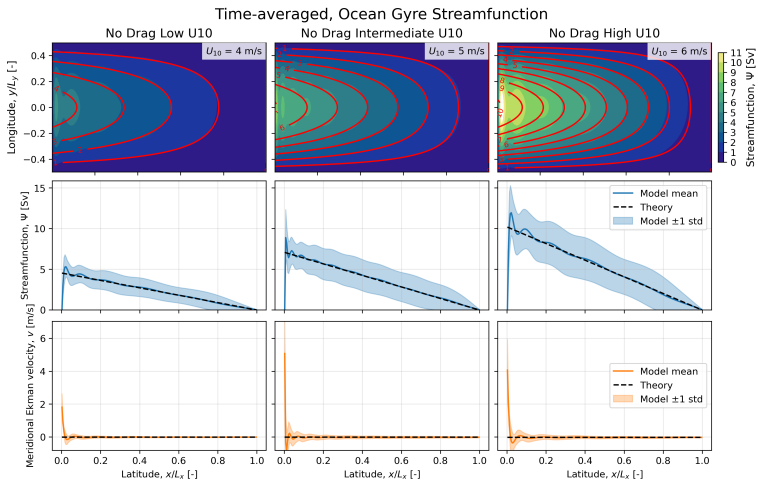


Figure: Streamfunction for the no-drag β -plane configuration. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

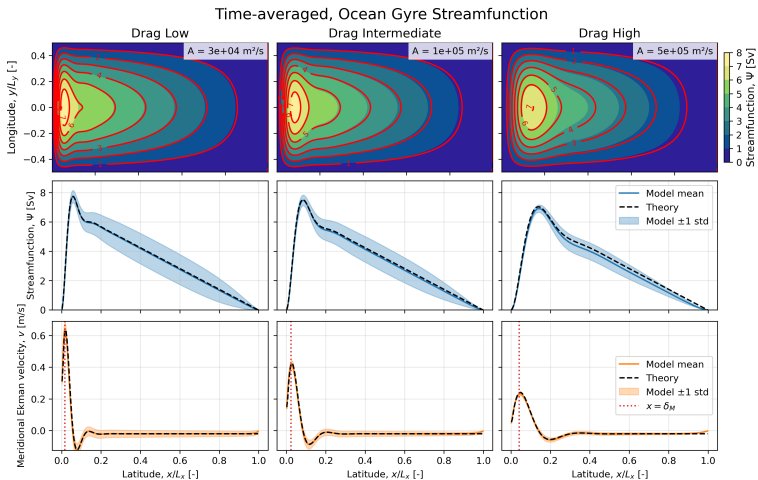


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Munk solution.

Result, The Munk solution

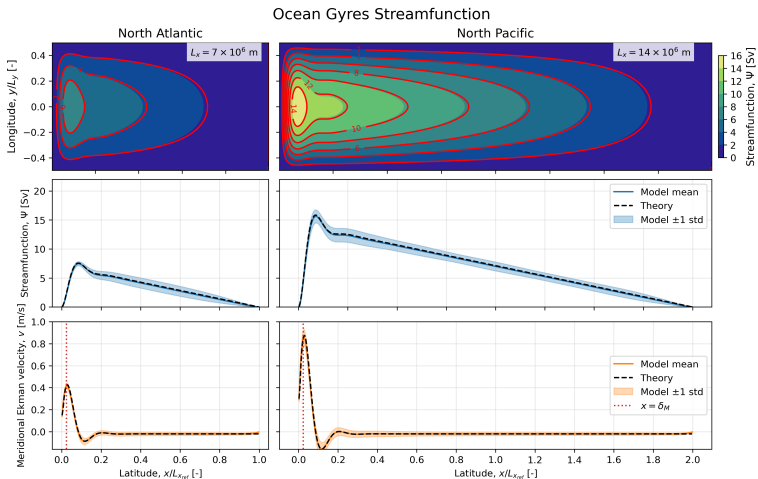


Figure: Streamfunction for the drag β -plane configuration with varying domain size. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

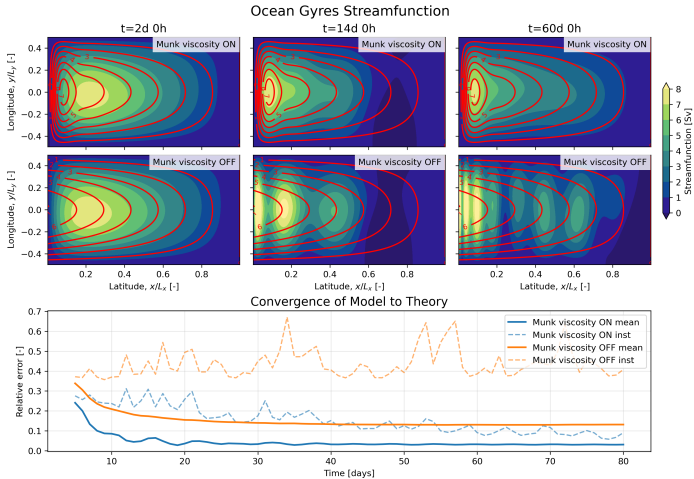


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Sverdrup and Munk solutions respectively.

Both numerical models show western boundary currents, but only Munk theory does. Why?

- The Sverdrup theory is linear, whereas the Munk theory is non-linear, and frictional force is opposite sign to the motion itself.
- The Munk viscosity creates a boundary layer where the frictional force can take care of mass conservation.
- Friction balances continuous input of energy. *Realistic?*
- Gives western boundary flow in right magnitude to real Gulf Stream ($\sim 2.5 \text{ m s}^{-1}$).

$$A \nabla^4 \psi - \beta \frac{\partial \psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau}$$

Three important conclusions:

- 1 Both numerical models showed western intensification, explained by Rossby wave propagation. But continuous energy input need dissipation to converge with time.
- 2 Only Munk theory showed a western boundary current, explained by frictional layer, although both numerical models showed return flow.
- 3 A larger domain gave same interior meridional velocity, which resulted in a stronger western boundary current.

References

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Extra figures

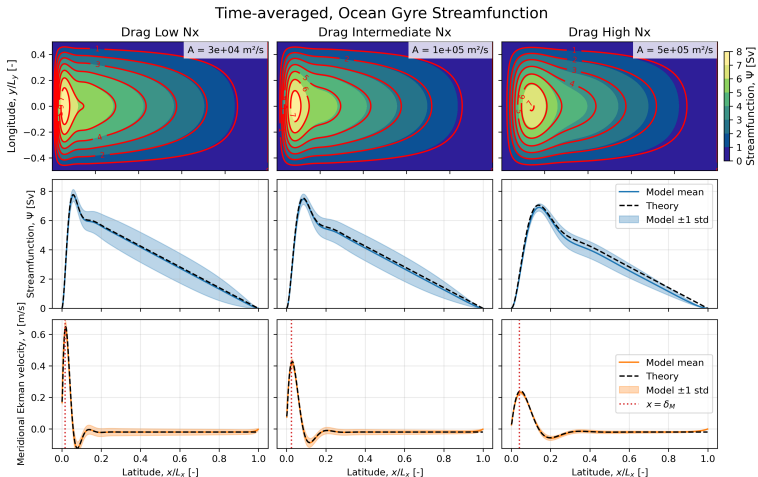


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Munk solution. This figure has twice as high resolution (i.e.

$$\Delta x = \Delta y = 17.5 \text{ km} \implies \frac{\delta_M}{\Delta x} \approx [6.6, 9.8, 16.8]$$